PSEUDO-COHERENCE DESCENDS FROM THE SMALL ETALE SITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: load-bearing boundedness input is the delivered lemma-epsilon-flat (L48088)

Editorial note. The label lemma-descend-pseudo-coherent names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Lemma lemma-epsilon-flat (delivered, L48088): the comparison functor epsilon^* is flat, identifying boundedness of E with boundedness of epsilon^*E
- Lemma lemma-descend-finite-type (in-source): finiteness of cohomology modules descends from the small etale site
- Cohomology on Sites, Lemmas sites-cohomology-lemma-pseudo-coherent-pullback, sites-cohomology-lemma-finite-cohomology, sites-cohomology-lemma-cone-pseudo-coherent

Source reference. Original equation 08H7: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 320–320. Prerequisite pointer invoked at 3140; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08HD: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 3081–3081. Prerequisite pointer invoked at 3174; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08H8: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 330–330. Prerequisite pointer invoked at 3161; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 08FT: sources/stacks-chow/tags.txt, source lines 10721–10721. Prerequisite pointer invoked at 3160; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 08FV: sources/stacks-chow/tags.txt, source lines 10723–10723. Prerequisite pointer invoked at 3185; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08FX: sources/stacks-chow/tags.txt, source lines 10725–10725. Prerequisite pointer invoked at 3172; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08H4: sources/stacks-chow/tags.txt, source lines 10767–10767. Prerequisite pointer invoked at 3146; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a scheme. Let E be an object of $D(\mathcal{O}_X)$. The following are equivalent*

(1) *E is m-pseudo-coherent, and*
(2) *ϵ^*E is m-pseudo-coherent on the small étale site of X.*

Here ϵ is as in (08H7).

Proof. The implication (1) $\Rightarrow$ (2) is a general fact, see Cohomology on Sites, Lemma 08H4. Assume ϵ^*E is m-pseudo-coherent. We will use without further mention that ϵ^* is an exact functor and that therefore

$$\epsilon^*H^i(E) = H^i(\epsilon^*E).$$

To show that E is m-pseudo-coherent we may work locally on X, hence we may assume that X is quasi-compact (for example affine). Since X is quasi-compact every étale covering $\{U_i \to X\}$ has a finite refinement. Thus we see that ϵ^*E is an object of $D^-(\mathcal{O}_{étale})$, see comments following Cohomology on Sites, Definition 08FT. By Lemma 08H8 it follows that E is an object of $D^-(\mathcal{O}_X)$.

Let $n \in \mathbf{Z}$ be the largest integer such that $H^n(E)$ is nonzero; then n is also the largest integer such that $H^n(\epsilon^*E)$ is nonzero. We will prove the lemma by induction on $n - m$. If $n < m$, then the lemma is clearly true. If $n \geq m$, then $H^n(\epsilon^*E)$ is a finite $\mathcal{O}_{étale}$-module, see Cohomology on Sites, Lemma 08FX. Hence $H^n(E)$ is a finite $\mathcal{O}_X$-module, see Lemma 08HD. After replacing X by the members of an open covering, we may assume there exists a surjection $\mathcal{O}_X^{\oplus t} \to H^n(E)$. We may locally on X lift this to a map of complexes $\alpha : \mathcal{O}_X^{\oplus t}[-n] \to E$ (details omitted). Choose a distinguished triangle

$$\mathcal{O}_X^{\oplus t}[-n] \to E \to C \to \mathcal{O}_X^{\oplus t}[-n+1]$$

Then C has vanishing cohomology in degrees $\geq n$. On the other hand, the complex ϵ^*C is m-pseudo-coherent, see Cohomology on Sites, Lemma 08FV. Hence by induction we see that C is m-pseudo-coherent. Applying Cohomology on Sites, Lemma 08FV once more we conclude. □